Surveillance Coverage, Clinical-Process Anchors, and Workload Constraints in Machine Learning Early Warning for Sepsis: An Operational Evaluation Audit

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Abstract

Objectives: To evaluate how surveillance protocol coverage, clinical-process anchor selection, and emitted-alert workload constraints govern the estimated utility and evaluability of machine learning early warning for sepsis.

Materials and Methods: We retrospectively evaluated 8 predictive architectures across 8,254 adult first ICU stays from MIMIC-III MetaVision (2008–2012; 97,700 hourly windows; prevalence 0.694%) under a second-day surveillance protocol (hours 19–30). We formalized an operational cascade ($S \to V \to O \to D$), evaluated outcome-stratified hybrid sets ($\mathcal{R}_0^{\text{hybrid}}, \mathcal{R}_2^{\text{hybrid}}$) as exploratory sensitivity analyses, quantified timestamp uncertainty for prescription orders, and estimated retrospective burden-matched operating characteristics under 12-hour alert suppression.

Results: Among 408 consensus Sepsis-3 stays, 257 (63.0%) developed onset after hour 36 ($V = 0$, median onset 49.0h). Among 151 surveillance-covered stays ($V = 1$, median onset 28.0h), 100 (66.2%) were first-bin assigned ($T_{\text{susp}} - T_{\text{S3}} \in [24\text{h}, 25\text{h})$), structurally ensuring pre-suspicion opportunity (150/151, 99.3%). Conversely, evaluating opportunity against earliest prescription orders under declared midnight point assignment eliminated all horizon-positive observations (0/151 definite; upper bound 47/151, 31.1%), rendering discrimination non-estimable for that task. At an alert budget of 10 episodes/100 patient-days (407 emitted alerts), the highest-performing model (BiMamba SSM) achieved an episode PPV of 1.72% (7/407, NNE 58.1), a covered sensitivity of 4.64% (7/151), and a total population yield of 1.72% (7/408). On the exploratory outcome-stratified hybrid set, discrimination remained stable (BiMamba: AUROC 0.6433 vs. 0.6326 canonical).

Discussion: Feasibility estimates depend heavily on whether process anchors are defined by clinical suspicion or prescription orders, and whether opportunity is structurally induced by onset search windows.

Conclusion: Sepsis predictive evaluations must transparently report protocol coverage, distinguish structurally induced opportunity from biological lead time, and benchmark under explicit alert workload constraints.

Keywords: Sepsis Early Warning, Surveillance Truncation, Anchor Selection, Alert Budget Frontier, State-Space Models.

1 Introduction

1.1 Background and Significance

Sepsis remains a leading cause of critical illness, inpatient mortality, and healthcare expenditure globally [1, 2]. Clinical treatment urgency depends critically on presentation: the Surviving Sepsis Campaign international guidelines recommend immediate antimicrobial therapy, ideally within one hour, for possible septic shock or a high likelihood of sepsis, while recommending rapid diagnostic assessment for possible sepsis without shock [3, 4]. To support prompt intervention, the biomedical informatics community has heavily invested in machine learning (ML) models for early warning. Scores of retrospective investigations have reported high discriminative capability, frequently reporting area under the receiver operating characteristic curve (AUROC) values between 0.80 and 0.88 ahead of consensus onset [5]. However, published ICU prediction studies differ substantially in outcome implementation, temporal alignment, and evaluation design; consequently, reported AUROCs should not be interpreted as directly comparable measures of deployment readiness [5, 6]. Prior MIMIC-III analyses have shown that alternative operational onset definitions can change measured performance as much as or more than changing model class [23]. These findings motivate separating reference-event construction from surveillance coverage and alert-policy evaluation. Time-sensitive sepsis benchmarks have also incorporated utility functions that reward and penalize alerts according to their timing [7]. Such approaches reinforce the need to report temporal evaluation rules, although benchmark utility is distinct from measured clinical benefit.

Clinical evaluations have reported heterogeneous performance and outcome associations [8, 9], while clinician-vignette studies have identified discrepancies between operational onset labels and recommended treatment timing [11]. External evaluation of the original Epic Sepsis Model demonstrated that reported performance depended heavily on the evaluation level and timing rules: hospitalization-level and prediction-level workload estimates were not interchangeable, emphasizing the need to report the exact unit of evaluation [8]. Prospective evaluations of early-warning systems such as TREWS have found associations between timely alert confirmation and improved outcomes among selected patients identified prior to antibiotic initiation [9, 10]. However, because provider response was observational rather than randomized, these findings support potential clinical utility without establishing universal causal efficacy [9].

Furthermore, ICU clinicians already encounter substantial alarm exposure from bedside physiological monitors [15]. In other clinical decision-support settings, repeated alerts and work complexity have been associated with lower alert acceptance, supporting explicit reporting of notification burden [16, 17].

Recent evaluations of sepsis early warning before clinical

intervention present divergent findings. Kamran et al. [12] reported a severe decline in Epic model AUROC (0.62 to 0.47) when excluding predictions following clinical recognition. Conversely, in a multi-center study spanning 198,494 encounters across nine hospitals, Dutta et al. [13] demonstrated that predictive performance was preserved under certain pre-clinical action definitions. Do et al. [14] further showed that continuous surveillance and alert silencing rules fundamentally alter clinical workload across health systems. This literature indicates that pre-intervention performance does not follow a uniform collapse, but rather depends on how onset labels, clinical-process anchors, and evaluation units are constructed.

These findings motivate the central question of this study: *Why do estimates of pre-clinical action opportunity and alert performance diverge across evaluation designs?* We show that existing evaluations frequently conflate three distinct operational mechanisms:

1. **Surveillance Protocol Coverage (V):** Fixed observation schedules may miss episodes occurring outside the evaluated interval.

2. **Clinical-Process Anchors and Timestamp Uncertainty (O):** Decision opportunities depend heavily on whether anchors are placed at clinical suspicion or recorded medication orders, and whether discrete search windows structurally induce pre-anchor separation.

3. **Emitted-Alert Workload Constraints (D):** Even when discrimination is preserved, operational alert budgets and suppression rules bound real-world sensitivity and positive predictive value (PPV).

To disentangle these mechanisms, we conducted an operational evaluation audit across 8 predictive architectures using electronic health record (EHR) data from the MIMIC-III database (v1.4; CareVue development and MetaVision evaluation replay) [20]. Our contributions are:

1. **Evaluation Audit Framework:** We formalize the operational cascade $S \rightarrow V \rightarrow O \rightarrow D$, disentangling surveillance protocol coverage, process-anchor eligibility, and emitted-alert workload.

2. **Structural Opportunity Discovery:** We demonstrate that pre-suspicion opportunity (150/151, 99.3%) is largely an algebraic consequence of the Sepsis-3 24h lookback window in first-bin cases ($T_{\text{susp}} - T_{\text{S3}} \in [24\text{h}, 25\text{h})$), rather than independent evidence of biological lead time.

3. **Operational Alert-Budget Frontiers:** We benchmark 8 models under strict alert budgets (5, 10, 20 episodes/100 patient-days) with a 12-hour alert suppression rule.

4. **Multi-Clock Timestamp Uncertainty Bounds:** We bound opportunity under order dates vs. interval bounds, showing why earliest-prescription proxies yield zero estimable positive cases under midnight assignment.

5. **Exploratory Outcome-Stratified Sensitivity:** We evaluate discrimination on hybrid sets ($\mathcal{R}_0^{\text{hybrid}}, \mathcal{R}_2^{\text{hybrid}}$) as an exploratory analysis, explaining why stability reflects outcome-specific censoring rather than uniform prospective policy.

2 Materials and Methods

2.1 Study Design and Cohort Definition

We conducted a retrospective evaluation with sequential prediction replay using the MIMIC-III critical care repository (v1.4) from BIDMC [20], evaluating transportability across clinical information systems from CareVue (2001–2008 development) to MetaVision (2008–2012 evaluation replay), with broad historical prescriptions providing a sensitivity analysis. Analyses used versioned extraction queries with dataset identifiers and checksums recorded in the release manifest. The study cohort was defined as adult patients ($\geq$ 18 years) admitted to an intensive care unit with a total ICU length of stay LOS $\geq$ 30h to evaluate mid-stay surveillance following baseline stabilization ($t = 0 \ldots 18$h). While conditioning on LOS $\geq$ 30h defines a restricted longer-stay ICU cohort (landmark design), it ensures all included patients remain under observation throughout the surveillance interval unless censored. To avoid within-patient clustering across multiple admissions, only the chronological first ICU stay per patient was retained. Complete cohort derivation schemas, exclusion criteria, train/validation/test partitions, feature dictionaries, missing-data rules, model architectures, and executable Sepsis-3 definitions are detailed in the Supplementary Methods Package.

Surveillance was evaluated under a prespecified second-day ICU surveillance protocol: predictions were generated at 12 scheduled hourly decision bins $[t-1, t]$ for $t \in \{19, 20, \ldots, 30\}$ hours post-admission. The test partition comprised 8,254 unique patient ICU stays (97,700 evaluated windows, 4,070.8 patient-days of surveillance volume). Stays were censored at Sepsis-3 onset (N = 110 stays, 623 windows) or upon observation-availability cessation prior to hour 30 under the prespecified feature-generation schedule (N = 180 control stays, 725 windows; all cohort stays had total LOS $\geq$ 30h). Reporting was organized in accordance with TRIPOD+AI and STROBE recommendations [21, 22].

2.2 Outcome Definition and Operational Clocks

Sepsis was defined according to a study-specific operational adaptation of consensus Sepsis-3 criteria [1, 2]: an acute increase in Sequential Organ Failure Assessment (SOFA) score $\geq$ 2 points over baseline. Baseline SOFA ($\text{SOFA}_{\text{base}}$) was defined as the minimum total SOFA score observed during the initial 24 hours of ICU stay ($t \in [0, 24\text{h}]$), representing the patient's baseline organ function prior to acute deterioration. Sepsis-3 onset was identified when total SOFA increased by $\geq$ 2 points above $\text{SOFA}_{\text{base}}$ within the clinical search window $[T_{\text{susp}} - 24\text{h}, T_{\text{susp}} + 12\text{h}]$ relative to suspected infection (defined as blood culture specimen acquisition paired with systemic antimicrobial prescription orders; detailed in Supplementary Section S1.5).

To resolve temporal precision and event ambiguity without conflating orders, intended start dates, and physical administrations, we track six operational milestones (with ICU admission as the reference origin $t = 0$, followed by five operational event clocks; Figure 1):

- $T_{\text{admit},i}$: ICU admission timestamp ($t = 0$, reference origin).

- $T_{\text{Rx,start},i}$: Earliest documented systemic antimicrobial prescription start date (`prescriptions.startdate`, calen-

dar date coarsened to midnight 00:00:00).

- $T_{\text{abx,pair},i}$: Systemic antimicrobial order date participating in the qualifying infection suspicion pair.
- $T_{\text{culture,pair},i}$: Blood culture specimen collection timestamp (`microbiologyevents.charttime`, minute-level precision).
- $T_{\text{susp},i}$: Operational infection-suspicion anchor, defined per Seymour et al. [2] as $\min(T_{\text{abx,pair}}, T_{\text{culture,pair}})$.
- $T_{\text{S3},i}$: Retrospective consensus Sepsis-3 onset timestamp.

When blood culture precedes antimicrobial prescription ($T_{\text{culture,pair}} < T_{\text{abx,pair}}$), $T_{\text{susp}} = T_{\text{culture,pair}} \neq T_{\text{abx,pair}}$. Let $h = 1$ hour and define the dimensionless lookback coordinate $x_i = (T_{\text{susp},i} - 24h - T_{\text{admit},i})/h$. The search window evaluates discrete hourly bins starting at baseline_start_h$_i = \lfloor x_i \rfloor$ and ending at window_end_h$_i = \lfloor (T_{\text{susp},i} + 12h - T_{\text{admit},i})/h \rfloor$. Because the upper boundary bin is floored, evaluated hours never exceed $T_{\text{susp}} + 12h$, strictly guaranteeing $T_{\text{susp},i} - T_{\text{S3},i} \geq -12.0h$ (empirical minimum: $-11.2h$). For episodes where acute ΔSOFA ≥ 2 is first observed at the earliest search bin ($N = 274/408$, 67.2% of all incident sepsis; and 100/151, 66.2% of surveillance-covered stays), onset is assigned as $T_{\text{S3},i} = T_{\text{admit},i} + h\lfloor x_i \rfloor$, yielding the operational lead gap:

$$G_i = T_{\text{susp},i} - T_{\text{S3},i} = 24h + h \ \text{frac}(x_i) \tag{1}$$

Because $0 \leq \text{frac}(x_i) < 1$ (dimensionless), first-bin-assigned episodes satisfy $24h \leq G_i < 25h$, with an analytical supremum of $25h$ (the reported $24.99h$ is the observed empirical maximum). In these first-bin episodes, pre-suspicion opportunity ($t < T_{\text{susp}}$) is algebraically guaranteed for any decision time evaluated prior to onset ($t < T_{\text{S3}} \approx T_{\text{susp}} - 24$h), and the 2-hour buffer is trivially satisfied. For the remaining 51 non-first-bin covered stays (33.8%), the median lead gap was 19.3h (IQR: 7.1–22.0h; 45 of 51 retaining pre-suspicion opportunity). Thus, high pre-suspicion opportunity in Sepsis-3 reflects the discrete lookback convention rather than independent empirical evidence of unprompted biological warning. Across the full 408 cohort, the median lead gap was +24.2h (IQR: 22.8h to 24.6h; range: -11.2h to +24.99h).

2.3 Operational Cascade and Outcome-Stratified Hybrid Risk Sets

Let C denote the evaluated test cohort of unique ICU stays ($N = 8,254$), and let $\mathcal{T}_i \subseteq \{19, 20, \ldots, 30\}$ denote the set of hourly decision bins evaluated for stay i. For patients who never develop consensus Sepsis-3 during hospitalization, we set $T_{\text{S3},i} = \infty$.

The full evaluated early-warning risk set is defined as:

$$\mathcal{R} = \left\{(i,t) : i \in C,\ t \in \mathcal{T}_i,\ t < T_{\text{S3},i}\right\} \tag{2}$$

The binary early-warning prediction target at anticipation horizon $H = 6$h is:

$$Y_i(t; H) = \mathbf{1}\left\{t < T_{\text{S3},i} \leq t + H\right\} \tag{3}$$

In the canonical unconstrained risk set ($\mathcal{R}_{\text{canonical}} = \mathcal{R}$), this definition yields exactly 97,700 evaluated hourly windows, comprising 678 horizon-positive observations ($Y = 1$) and 97,022 negative observations ($Y = 0$, prevalence $\pi = 0.694\%$).

To evaluate restricted decision windows, we define the outcome-stratified hybrid risk sets relative to reference clock C_i:

$$\mathcal{R}_{\delta}^{\text{hybrid}} = \left\{(i,t) \in \mathcal{R} : t + \delta \leq C_i\right\} \tag{4}$$

where $C_i = T_{\text{susp},i} = \min(T_{\text{culture,pair},i}, T_{\text{abx,pair},i})$ for incident sepsis stays ($S_i = 1$; determined by $T_{\text{abx,pair}}$ in 399 stays and blood culture charttime in 9 stays), and $C_i = T_{\text{Rx,start},i}$ for non-sepsis control stays ($S_i = 0$) if systemic antimicrobials were ordered during hospitalization ($C_i = \infty$ otherwise). Exactly 2,287 control stays had $T_{\text{Rx,start}} \leq 19$h, censoring all their windows ($W_{\text{fully excluded}} = 27,388$). Among 83 later-prescription control stays ($T_{\text{Rx,start}} > 19$h), 113 windows were censored post-prescription at $\delta = 0$h ($65,926 = 93,427 - 27,388 - 113$), and 49 additional windows were removed by the 2-hour proxy-timing margin ($65,877 = 65,926 - 49$). We evaluated $\mathcal{R}_0^{\text{hybrid}}$ ($\delta = 0$h; 70,192 windows across 5,967 stays; prevalence 0.960%) and $\mathcal{R}_2^{\text{hybrid}}$ ($\delta = 2$h; 70,129 windows across 5,967 stays; prevalence 0.945%).

Importantly, sepsis cases and controls were restricted using different clinical-process anchors. Differences from canonical discrimination therefore reflect both changes in the evaluated observations and the outcome-specific selection rule; they are not estimates of performance under a uniform prospective eligibility policy. Within this restricted surveillance protocol and under declared midnight assignment of prescription-start dates, no horizon-positive observations remained under uniform earliest-prescription eligibility ($\mathcal{R}_0^{\text{Rx}}$: 5,559 control stays, 65,926 negative windows, 0 positive windows, $N_{\text{total}} = 65,926$; $\mathcal{R}_2^{\text{Rx}}$: 65,877 negative windows; $N_+^{\text{Rx}} = 0$; prevalence strictly 0.000%). Discrimination was therefore not meaningfully estimable: with no eligible positive observations, AUROC was undefined and AUPRC did not provide an informative discrimination estimate. PPV would be zero for a policy emitting alerts in this all-negative evaluated set, and undefined if no alerts were emitted.

2.4 Alert-Budget Operating Frontiers and Refractory Policy

To quantify simulated emitted-alert burden, we evaluated fixed alert budgets $q \in \{5.0, 10.0, 20.0\}$ emitted alert episodes per 100 surveillance patient-days. Under a 12-hour suppression rule, each stay emits at most one alert episode within the evaluated 12-hour interval ($N_{\text{episodes},i} \leq 1$). We explicitly report three distinct operational metrics with transparent denominators:

$$\text{Episode PPV} = \frac{N_{\text{timely detections}}}{N_{\text{emitted alerts}}} \tag{5}$$

$$\text{Covered Sensitivity} = \frac{N_{\text{timely detections}}}{N_{\text{covered sepsis cases}}} \quad (N = 151) \tag{6}$$

$$\text{Population Yield} = \frac{N_{\text{timely detections}}}{N_{\text{all target sepsis cases}}} \quad (N = 408) \tag{7}$$

At an alert budget of 10 episodes per 100 patient-days ($N_{\text{emitted alerts}} = 407$), the numerical coincidence between Episode PPV ($7/407 = 1.72\%$) and Population Yield ($7/408 = 1.72\%$) arises strictly from the proximity of total emitted alerts to total target sepsis cases; their conceptual definitions and denominators are distinct.

We estimated retrospective burden-matched operating characteristics by selecting score cutoffs from the held-out score

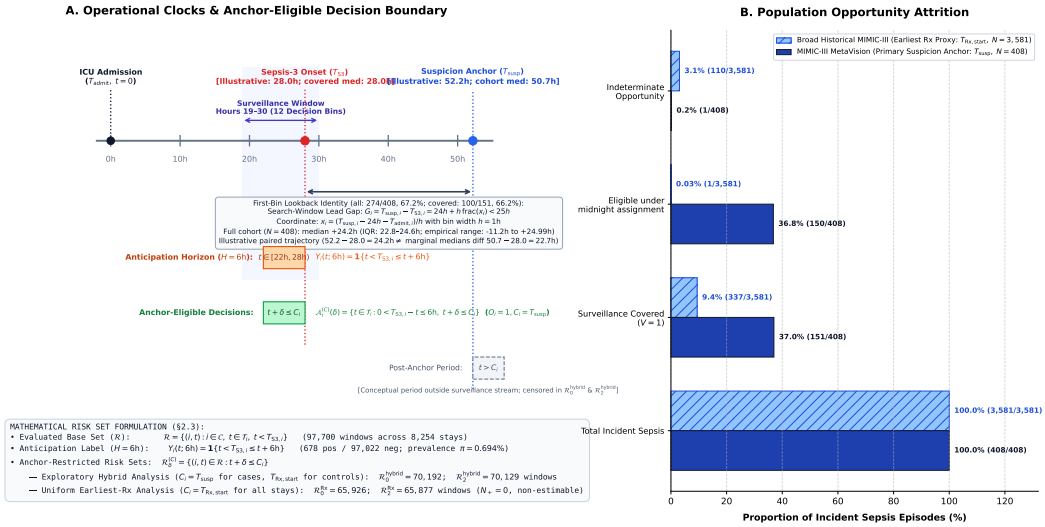


Figure 1: Operational Clocks and Anchor-Dependent Evaluation Eligibility in the MIMIC-III MetaVision Cohort. The timeline is schematic; displayed population summaries do not represent a single paired patient trajectory. Eligibility relative to a retrospective suspicion anchor does not establish absence of prior antimicrobial exposure. Panel A depicts the clinical timeline for an ICU-incident sepsis stay, illustrating the discrete search-window lead $G_i = 24h + h\ \mathrm{frac}(x_i) < 25h$ for first-bin-assigned episodes ($N = 274/408$, 67.2%; median +24.2h; empirical max 24.99h), the anticipation horizon $t \in [22\mathrm{h}, 28\mathrm{h})$, anchor-eligible decision times $\mathcal{A}_i^{(C)}(\delta)$, and the conceptual post-anchor period. Actual model evaluations occur at discrete scored timestamps $t \in \mathcal{T}_i$. Panel B displays population opportunity attrition comparing Broad Historical MIMIC-III (earliest prescription proxy $T_{\mathrm{Rx,start}}$, $N = 3,581$) and MIMIC-III MetaVision (suspicion anchor T_{susp}, $N = 408$), reporting exact stay counts beside each opportunity tier under declared midnight assignment. *Alt-text:* Two-panel diagram displaying schematic timeline clocks and formal risk-set definitions in Panel A, and horizontal bar charts comparing surveillance coverage and treatment-precedence opportunity in Panel B.

distribution. These operating points describe empirical performance at matched emitted-alert rates; they do not represent thresholds prospectively fixed for online deployment. Patient selection used the maximum score over the evaluated interval, whereas detection timing was assessed at the first eligible threshold crossing under the complete tie-selection rule:

$$t_{\mathrm{first},i} = \min\{t \in \mathcal{T}_i : \mathrm{Elig}_i(t; \tau, \mathcal{B}) = 1\} \tag{8}$$

where $\mathcal{B}$ denotes the deterministic prioritization rule:

$$\mathrm{Elig}_i(t; \tau, \mathcal{B}) = \mathbf{1}\{i \in \mathcal{S}_{\mathrm{alerted}}(K) \ \wedge \ s_i(t) \geq \tau\} \tag{9}$$

with $\mathcal{S}_{\mathrm{alerted}}(K)$ defined as the top K patient replicas under the lexicographical sort:

$$\left(-\max_t s_i(t),\ \mathrm{ICUSTAY_ID},\ \mathrm{copy_idx}\right)$$

2.5 Predictive Models and Statistical Inference

We evaluated 8 models spanning three major architectural paradigms:

1. **State-Space Models:** Bidirectional Mamba (BiMamba SSM) utilizing linear-time selective state-space layers [18, 19].
2. **Deep Sequence Transformers:** Mask-Aware Clinical Transformer (incorporating explicit missingness masks) and Baseline Unmasked Transformer.
3. **Tabular and Ensemble Baselines:** Random Forest, XGBoost, LightGBM, Logistic Regression (L2), and a Multi-Model Stacking Ensemble.

Models consumed a causal 20-hour observation lookback $[t - 20\mathrm{h}, t)$ across 21 physiological time series. All models were trained on fixed pre-surveillance development partitions with hyperparameters and feature normalization frozen prior to this operational evaluation. Complete architecture details, hyperparameter grids, and the stacking protocol are detailed in the Supplementary Methods Package.

Uncertainty was quantified using patient-clustered paired bootstrapping ($B = 1,000$ resamples). For Procedure B (Empirical Burden-Matched Bootstrap), exposure $E_b = \frac{1}{24}\sum_{i \in b} h_i$ and alert target $K_b = \mathrm{round}(q \cdot E_b/100)$ were dynamically recomputed within each resample. NNE confidence intervals were derived by exact reciprocal transformation of unrounded PPV percentiles: $\mathrm{NNE}_{\mathrm{low}} = 100/\mathrm{PPV}_{\mathrm{upper}}$ and $\mathrm{NNE}_{\mathrm{high}} = 100/\mathrm{PPV}_{\mathrm{low}}$ (with $+\infty$ when $\mathrm{PPV}_{\mathrm{low}} = 0$).

3 Results

3.1 Cohort Progression and Denominator Accounting

Table 1 traces all 8,254 ICU admissions through surveillance eligibility, onset timing, lookback horizons, and pre-treatment actionability. Across all 408 consensus Sepsis-3 stays, median onset was 49.0h post-admission. Exactly 257 stays (63.0%; median onset: 69.0h) had clinical onset after hour 36. Under the specified observation schedule and 6-hour horizon, these 257 stays had no evaluated horizon-positive decision time ($V = 0$). The remaining 151 stays (37.0%; median onset: 28.0h) had active protocol surveillance coverage ($V = 1$). Of the 151 covered stays, 150 had at least one horizon-positive decision time at or before recorded treatment initiation (99.3% within-covered opportunity; 36.8% of total incident sepsis), providing pre-proxy opportunity under midnight assignment ($\mathcal{R}_0$).

Table 1: Cohort Progression and Denominator Accounting Ledger. Denominators traced from the base test cohort through surveillance eligibility, sepsis onset timing, prediction horizons, and pre-treatment actionability.

Progression Stage / Cohort Partition	Stays (N)	Evaluated Windows	Surveillance Volume	Prevalence	Clinical Definition & Measurement Convention
1. Base Prospective-Replay Test Cohort	8,254	97,700 / 99,048 sched.	4,070.8 pt-days	100.0%	Adult first ICU stays ≥ 30h; 1,348 censored (623 onset, 725 cessation)
2. Non-Sepsis Control ICU Stays	7,846	93,427 neg. windows	3,892.8 pt-days	95.06%	Stays without consensus Sepsis-3 during hospitalization
3. Consensus Sepsis-3 Stays ($S = 1$)	408	4,273 (678 pos, 3,595 neg)	178.0 pt-days	4.94%	Blood culture + antibiotic + acute ΔSOFA ≥ 2; full cohort median onset 49.0h
3a. Late-Onset Sepsis (Onset > 36h)	257	3,084 neg. windows	128.5 pt-days	63.0% of septic	Onset > 36h (median: 69.0h); lookback $[T_{S3} - 6, T_{S3}]$ outside [19, 30] ($V = 0$)
3b. Surveillance-Covered ($V = 1$)	151	1,189 (678 pos, 511 neg)	49.5 pt-days	37.0% of septic	Onset in [20h, 36h] (median: 28.0h); lookback intersects window ($V = 1$)
4. Canonical Sepsis-3 Risk Set ($H = 6$h)	8,254	97,700 (678 pos / 97,022 neg)	4,070.8 pt-days	0.694%	Full unconstrained risk set ($t \in \mathcal{T}_i,\ t < T_{S3,i}$)
5. Outcome-Stratified Hybrid Set $\mathcal{R}_0^{\text{hybrid}}$ ($\delta = 0$h)	5,967	70,192 (674 pos / 69,518 neg)	2,924.7 pt-days	0.960%	Censoring post-anchor ($t \le C_i$; T_{susp} cases, $T_{\text{Rx,start}}$ controls); 150/151 covered eligible
6. Outcome-Stratified Hybrid Set $\mathcal{R}_2^{\text{hybrid}}$ ($\delta = 2$h)	5,967	70,129 (663 pos / 69,466 neg)	2,922.0 pt-days	0.945%	2-hour proxy-timing margin ($t + 2\text{h} \le C_i$); 149/151 covered eligible

3.2 Outcome-Stratified Hybrid Discrimination Analysis

Table 2 presents discrimination across Canonical Sepsis-3, Hybrid Set ($\mathcal{R}_0^{\text{hybrid}}$), and Hybrid Set with 2h Buffer ($\mathcal{R}_2^{\text{hybrid}}$). Discrimination changed little in the outcome-stratified hybrid analysis (BiMamba SSM: AUROC 0.6326 canonical vs. 0.6433 on $\mathcal{R}_0^{\text{hybrid}}$, paired ΔAUROC = +0.0107, 95% CI: +0.0060 to +0.0175; Stacking Ensemble: 0.6284 vs. 0.6351, paired ΔAUROC = +0.0067). Because cases and controls were restricted using different anchors, this finding does not establish preserved discrimination under a uniform pre-suspicion or pre-prescription policy. The increase in AUPRC (BiMamba: 1.209% to 1.656%) coincided with higher positive prevalence (0.694% vs. 0.960%) and changes in the evaluated case mix; these contributions were not separately identified. These results are directionally consistent with the antibiotic-order sensitivity findings of Dutta et al. (2026) [13].

3.3 Observed Burden–Detection Trade-Off Under This Protocol

Table 3 and Figure 2 demonstrate the observed trade-off under the prespecified alert-budget scenarios: bedside detection yield is constrained by alert-rate limits. Alerts were selected across the full canonical surveillance stream (4,070.8 patient-days; 97,700 windows across all 8,254 test stays); C_i was used only retrospectively to adjudicate advance-warning timing ($t_{\text{first},i} + \delta \le T_{\text{susp},i}$ for sepsis cases), rather than excluding observations prior to alert selection.

At an emitted workload of 10 alert episodes per 100 patient-days (407 emitted episodes across 4,070.8 patient-days):

- Across all 8 models, episode-level PPV ranged from 0.98% to 1.72%, corresponding to 58.1 to 101.8 simulated evaluations per detected patient (NNE).
- BiMamba SSM and LightGBM each uniquely detected **7 patients** (PPV = 1.72%, 95% CI: 0.49%–2.95%; NNE = 58.1, 95% CI: 33.9–203.5).
- Stacking Ensemble, XGBoost, and Random Forest each detected **6 patients** (PPV = 1.47%, NNE = 67.8).
- In this observed evaluation, the count of horizon-correct emitted alerts and the count of alerts additionally satisfying the two-hour pre-suspicion proxy-timing margin ($t_{\text{first},i} + 2\text{h} \le T_{\text{susp},i}$) were identical across all models and operating points (e.g., 7 of 7 detected patients for BiMamba SSM and LightGBM at 10 ep/100 pt-d; equality in sample: $N_{\text{det}} = N_{\text{det,pre-susp}}$). These detections occurred before the selected retrospective suspicion anchor (T_{susp}), not necessarily before earliest antimicrobial prescribing ($T_{\text{Rx,start}}$).
- Model rankings varied across budgets: at 20 episodes/100 pt-days, XGBoost detected 15 patients (NNE = 54.3) and Mask-Aware Transformer detected 14 patients (NNE = 58.1), highlighting workload-dependent operating behavior.

3.4 Two-Axis Subgroup and Timestamp Uncertainty Matrix

Table 4 presents the cross-tabulation of Sepsis Onset Timing against Treatment Proxy Timing. Because MIMIC-III prescription start dates (`prescriptions.startdate`) lack time of day, order initiation is evaluated both under declared point assignment (midnight a_i) and under calendar-day intervals $[a_i, b_i)$ = [midnight, next midnight).

Opportunity depended strongly on the selected clinical-process anchor. Under the primary benchmark evaluating decision opportunity relative to the infection-suspicion anchor (T_{susp}), 150 of 151 covered patients (99.3% within-covered; 36.8% of total incident sepsis) had decision bins preceding T_{susp} under declared midnight point assignment a_i. Because $a_i \le T_{\text{susp},i}$ across $[a_i, b_i)$, these 150 stays are definitely eligible throughout the calendar-day interval ($O_i^{\text{def}} = 150$), with only 1 indeterminate stay ($O_i^{\text{indet}} = 1$), bounding opportunity between 36.8% and 37.0% ($[150, 151]/408$; [99.3%, 100.0%] within-covered). All indicators strictly satisfy $O_i^{\text{def}} \le O_i^{\text{point}} \le O_i^{\text{poss}}$ and $O_i^{\text{poss}} = O_i^{\text{def}} + O_i^{\text{indet}}$.

In contrast, opportunity was substantially lower and uncertain relative to the earliest hospital prescription-start proxy ($T_{\text{Rx,start}}$): exactly 0 covered stays had definite opportunity under midnight point-assignment ($O_i^{\text{def}} = 0, O_i^{\text{point}} = 0$), with 47 indeterminate stays ($O_i^{\text{indet}} = 47$), establishing bounds [0.0%, 11.5%]

Table 2: Benchmark Discrimination Across Outcome-Stratified Hybrid Risk Sets (Held-Out Point Estimates and Clustered 95% CIs). Point estimates calculated directly on held-out test data ($B = 0$); 95% CIs derived from $B = 1,000$ patient-clustered bootstrap resamples. Evaluated on hybrid risk sets $\mathcal{R}_0^{\text{hybrid}}$ and $\mathcal{R}_2^{\text{hybrid}}$ censoring cases at infection-suspicion anchor T_{susp} and controls at earliest prescription $T_{\text{Rx,start}}$. Sepsis cases and controls were restricted using different clinical-process anchors; differences from canonical discrimination reflect both evaluated observations and outcome-specific selection, and do not represent a uniform prospective eligibility policy.

Model Architecture / Metric	Canonical Sepsis-3 ($H = 6$h)	Hybrid $\mathcal{R}_0^{\text{hybrid}}$ ($\delta = 0$h)	Hybrid $\mathcal{R}_2^{\text{hybrid}}$ ($\delta = 2$h)	Paired ΔAUROC ($\mathcal{R}_0^{\text{hybrid}}$ − Canon)
BiMamba SSM	0.6326 (0.5872–0.6733)	**0.6433** (0.5998–0.6835)	**0.6447** (0.6002–0.6860)	**+0.0107** (+0.0060 to +0.0175)
AUPRC	*1.209% (0.892–1.769)*	***1.656%*** *(1.235–2.329)*	***1.647%*** *(1.220–2.331)*	—
Multi-Model Stacking Ensemble	0.6284 (0.5829–0.6699)	0.6351 (0.5897–0.6771)	0.6370 (0.5913–0.6799)	+0.0067 (+0.0021 to +0.0137)
AUPRC	*1.206% (0.882–1.748)*	*1.621% (1.207–2.263)*	*1.613% (1.198–2.255)*	—
Random Forest	0.6108 (0.5643–0.6535)	0.6143 (0.5684–0.6575)	0.6173 (0.5712–0.6612)	+0.0034 (-0.0008 to +0.0101)
AUPRC	*1.080% (0.821–1.434)*	*1.470% (1.138–1.920)*	*1.461% (1.120–1.916)*	—
Mask-Aware Clinical Transformer	0.5942 (0.5467–0.6410)	0.5961 (0.5490–0.6422)	0.5954 (0.5480–0.6426)	+0.0020 (-0.0024 to +0.0083)
AUPRC	*1.013% (0.766–1.303)*	*1.382% (1.048–1.760)*	*1.365% (1.032–1.752)*	—
Tabular GBDT (XGBoost)	0.5670 (0.5208–0.6102)	0.5643 (0.5186–0.6082)	0.5652 (0.5192–0.6099)	-0.0027 (-0.0070 to +0.0038)
AUPRC	*0.977% (0.739–1.357)*	*1.320% (1.008–1.817)*	*1.313% (0.998–1.811)*	—
Tabular GBDT (LightGBM)	0.5613 (0.5161–0.6026)	0.5587 (0.5139–0.6010)	0.5598 (0.5145–0.6033)	-0.0026 (-0.0070 to +0.0038)
AUPRC	*0.893% (0.694–1.195)*	*1.222% (0.957–1.629)*	*1.214% (0.949–1.630)*	—
Logistic Regression (L2)	0.5593 (0.5097–0.6048)	0.5654 (0.5157–0.6107)	0.5657 (0.5150–0.6127)	+0.0061 (+0.0023 to +0.0116)
AUPRC	*0.939% (0.716–1.218)*	*1.337% (1.012–1.740)*	*1.325% (1.002–1.731)*	—
Clinical Transformer (Baseline)	0.5239 (0.4808–0.5653)	0.5296 (0.4851–0.5712)	0.5286 (0.4834–0.5705)	+0.0057 (+0.0019 to +0.0110)
AUPRC	*0.808% (0.609–1.381)*	*1.147% (0.857–1.904)*	*1.131% (0.845–1.894)*	—

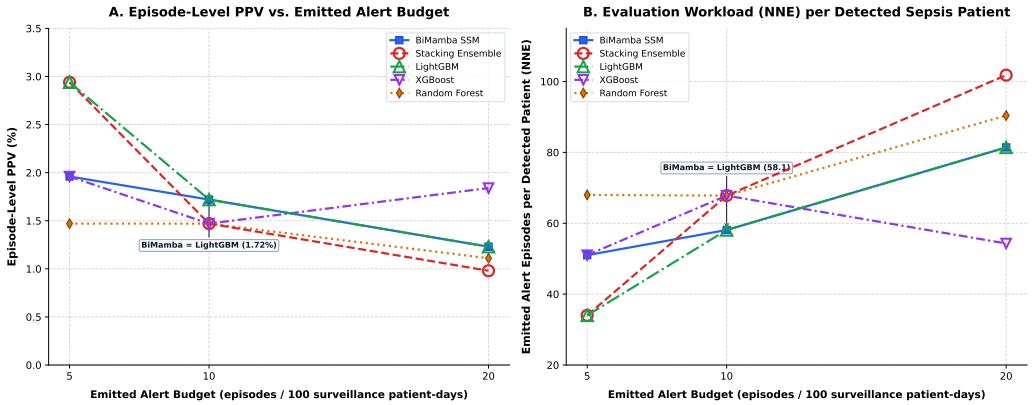


Figure 2: Operational Alert-Budget Performance Frontier. Panel A displays episode-level positive predictive value (PPV %) across prespecified alert budgets. Panel B displays the Number Needed to Evaluate (NNE) per detected patient under a 12-hour alert suppression rule. Lines connect discrete prespecified operating budgets (5, 10, and 20 emitted episodes per 100 patient-days) for a representative 5-model subset; complete 8-model results, bootstrap confidence intervals, and boundary-degenerate notes are reported in Table 3. BiMamba SSM and LightGBM have coincident operating points at budgets of 10 and 20 episodes per 100 surveillance patient-days. *Alt-text:* Two-panel plot showing PPV (Panel A) and evaluation workload per detected patient (Panel B) across alert budgets of 5, 10, and 20 episodes/100 patient-days for 5 models under 12h alert suppression.

$([0, 47]/408;\ [0.0\%, 31.1\%]$ within-covered). Because no horizon-positive cases remained before point-assigned earliest prescriptions, this task was not evaluable for discrimination under midnight point assignment. In broad historical MIMIC-III ($N = 3,581$), intervals generate 1 definitely eligible stay, 110 indeterminate stays, and bounds $[0.03\%, 3.10\%]$. The 150/151 result under T_{susp} does not establish that patients were unexposed to antimicrobials or that models predicted prior to first administration. Equally, zero definite opportunity under $T_{\text{Rx,start}}$ reflects date coarsening, not verified bedside exposure.

3.5 Cumulative Population Coverage Decomposition

Figure 3 illustrates the cumulative population coverage decomposition:

$$\frac{151}{408} \times \frac{150}{151} \times \frac{7}{150} = \frac{7}{408} = 1.72\%$$

1. **Surveillance Truncation** ($151/408 = 37.0\%$)**:** 63.0% of sepsis patients develop onset after hour 36, outside the prespecified second-day surveillance protocol.
2. **Suspicion-Anchor Eligibility** ($150/151 = 99.3\%$)**:** Of the 151 covered stays, 150 had at least one horizon-positive decision time at or before the retrospective suspicion anchor (T_{susp}) [149 of 151 (98.7%) under a 2-hour pre-suspicion margin].
3. **Model-Alert Policy Detection** ($7/150 = 4.67\%$)**:** Under an operational alert policy combining score ranking, a 10 episodes/100 patient-days budget, and a 12-hour alert suppression rule, BiMamba SSM detects 7 patients (1.72% whole-population yield; NNE = 58.1).

4 Discussion

This investigation characterizes the operational trade-offs governing machine learning early warning for sepsis in ICU surveillance:

1. **Outcome-Dependent Anchor Selection Governs Evaluability:** The canonical surveillance analysis quantified

Table 3: Operational Burden-Matched Alert Frontiers Under 12h Refractory Suppression. Evaluated under Procedure B with exposure recomputation and first-emitted-alert timing verification.

Budget	Model Architecture	Cutoff (τ)	Alerts (K)	Episode PPV % [95% CI]	Det (N)	Sens. Cov %	Yield %	NNE [95% CI]
5.0 ep / 100 pt-d	Stacking Ensemble	0.0145	204	2.94 (0.98–5.88)%	6	3.97 (1.31–7.65)%	1.47 (0.49–2.86)%	34.0 (17.0–102.0)
	LightGBM	0.0117	204	2.94 (0.98–5.39)%	6	3.97 (1.26–7.10)%	1.47 (0.47–2.61)%	34.0 (18.5–102.0)
	BiMamba SSM	0.0187	204	1.96 (0.49–4.45)%	4	2.65 (0.65–6.20)%	0.98 (0.24–2.31)%	51.0 (22.5–204.0)
	XGBoost	0.0122	204	1.96 (0.49–3.94)%	4	2.65 (0.63–5.63)%	0.98 (0.23–2.09)%	51.0 (25.4–204.0)
	Random Forest	0.0177	204	1.47 (0.00–3.45)%	3	1.99 (0.00–4.49)%	0.74 (0.00–1.70)%	68.0 (29.0–∞)
	Clinical Transformer	0.0101	204	0.98 (0.00–2.46)%	2	1.32 (0.00–3.50)%	0.49 (0.00–1.27)%	102.0 (40.6–∞)
	Logistic Regression	0.0103	204	0.98 (0.00–2.94)%	2	1.32 (0.00–3.85)%	0.49 (0.00–1.45)%	102.0 (34.0–∞)
	Mask-Aware Transformer	0.0120	204	0.00 (0.00–0.00)%*	0	0.00 (0.00–0.00)%*	0.00 (0.00–0.00)%*	Undefined (∞)
10.0 ep / 100 pt-d	BiMamba SSM	0.0165	407	1.72 (0.49–2.95)%	7	4.64 (1.33–8.16)%	1.72 (0.49–2.93)%	58.1 (33.9–203.5)
	LightGBM	0.0110	407	1.72 (0.49–2.95)%	7	4.64 (1.38–7.74)%	1.72 (0.49–2.89)%	58.1 (33.9–203.5)
	Stacking Ensemble	0.0131	407	1.47 (0.25–2.46)%	6	3.97 (0.70–6.98)%	1.47 (0.26–2.54)%	67.8 (40.7–407.0)
	XGBoost	0.0114	407	1.47 (0.49–2.94)%	6	3.97 (1.32–7.65)%	1.47 (0.49–2.85)%	67.8 (34.0–203.5)
	Random Forest	0.0158	407	1.47 (0.49–2.70)%	6	3.97 (1.26–7.29)%	1.47 (0.47–2.70)%	67.8 (37.0–204.0)
	Mask-Aware Transformer	0.0110	407	1.23 (0.25–2.46)%	5	3.31 (0.74–6.67)%	1.23 (0.26–2.48)%	81.4 (40.7–407.0)
	Clinical Transformer	0.0094	407	1.23 (0.25–2.46)%	5	3.31 (0.68–6.38)%	1.23 (0.25–2.37)%	81.4 (40.7–407.0)
	Logistic Regression	0.0095	407	0.98 (0.25–1.97)%	4	2.65 (0.59–5.51)%	0.98 (0.23–2.07)%	101.8 (50.8–408.0)
20.0 ep / 100 pt-d	XGBoost	0.0105	814	1.84 (0.98–2.70)%	15	9.93 (5.26–14.47)%	3.68 (1.87–5.29)%	54.3 (37.0–101.9)
	Mask-Aware Transformer	0.0099	814	1.72 (0.74–2.58)%	14	9.27 (4.54–13.77)%	3.43 (1.59–5.19)%	58.1 (38.8–135.7)
	Logistic Regression	0.0088	814	1.35 (0.49–2.09)%	11	7.28 (2.74–11.11)%	2.70 (0.99–4.18)%	74.0 (47.8–203.7)
	BiMamba SSM	0.0142	814	1.23 (0.49–2.09)%	10	6.62 (3.07–10.89)%	2.45 (1.05–4.10)%	81.4 (47.9–203.2)
	LightGBM	0.0105	814	1.23 (0.61–2.09)%	10	6.62 (3.11–11.04)%	2.45 (1.15–4.10)%	81.4 (47.9–163.0)
	Random Forest	0.0131	814	1.11 (0.49–1.85)%	9	5.96 (2.70–9.68)%	2.21 (0.96–3.69)%	90.4 (54.2–203.7)
	Stacking Ensemble	0.0115	814	0.98 (0.37–1.72)%	8	5.30 (2.04–9.15)%	1.96 (0.74–3.43)%	101.8 (58.1–271.3)
	Clinical Transformer	0.0088	814	0.86 (0.25–1.48)%	7	4.64 (1.51–8.18)%	1.72 (0.52–3.04)%	116.3 (67.8–407.0)

*At zero observed detections, the empirical percentile-bootstrap interval was degenerate at zero. This does not establish zero population detection probability. NNE intervals derived by exact reciprocal transformation. In this observed sample, horizon-correct detections and detections satisfying the two-hour pre-suspicion proxy-timing margin ($t_{\text{first},i} + 2\text{h} \leq T_{\text{susp},i}$) were identical across all models and operating points.

Table 4: Two-Axis Subgroup and Timestamp Uncertainty Matrix. Opportunity bounds conditional on fixed consensus Sepsis-3 timestamps and surveillance coverage.

Cohort & Timestamp Source	Sepsis Onset Category	Total (N)	Covered ($V = 1$)	Def. Elig (O^{def})	Point (O^{point})	Indet (O^{indet})	Poss (O^{poss})	Subgroup Bounds	Contribution to Cohort
MetaVision Prospective-Replay	Early Incident (≤ 24h)	47	47 (100.0%)	47	47	0	47	[100.0%, 100.0%]	[47, 47] / 408 (11.5%)
(Primary Benchmark: Suspicion Anchor T_{susp})	Intermediate (24–48h)	156	104 (66.7%)	103	103	1	104	[66.0%, 66.7%]	[103, 104] / 408 (25.2–25.5%)
	Late Incident (> 48h)	205	0 (0.0%)	0	0	0	0	[0.0%, 0.0%]	0 / 408 (0.0%)
	All Incident Sepsis-3	**408**	**151 (37.0%)**	**150**	**150**	**1**	**151**	**[36.8%, 37.0%]**	**[150, 151] / 408 (36.8–37.0%)**
MetaVision Sensitivity: Earliest Rx	Early Incident (≤ 24h)	47	47 (100.0%)	0	0	38	38	[0.0%, 80.9%]	[0, 38] / 408 (0.0–9.3%)
(Pre-Exposure Sensitivity: Earliest Rx $T_{\text{Rx,start}}$)	Intermediate (24–48h)	156	104 (66.7%)	0	0	9	9	[0.0%, 5.8%]	[0, 9] / 408 (0.0–2.2%)
	Late Incident (> 48h)	205	0 (0.0%)	0	0	0	0	[0.0%, 0.0%]	0 / 408 (0.0%)
	All Incident Sepsis-3	**408**	**151 (37.0%)**	**0**	**0**	**47**	**47**	**[0.0%, 11.5%]**	**[0, 47] / 408 (0.0–11.5%)**
Broad Historical MIMIC-III	Early Incident (≤ 24h)	2,641	102 (3.9%)	0	0	74	74	[0.00%, 2.80%]	[0, 74] / 3,581 (0.00–2.07%)
(Date-Level Proxy: Earliest Rx $T_{\text{Rx,start}}$)	Intermediate (24–48h)	361	235 (65.1%)	1	1	36	37	[0.28%, 10.25%]	[1, 37] / 3,581 (0.03–1.03%)
	Late Incident (> 48h)	579	0 (0.0%)	0	0	0	0	[0.00%, 0.00%]	[0, 0] / 3,581 (0.00%)
	All Incident Sepsis-3	**3,581**	**337 (9.4%)**	**1**	**1**	**110**	**111**	**[0.03%, 3.10%]**	**[1, 111] / 3,581 (0.03–3.10%)**

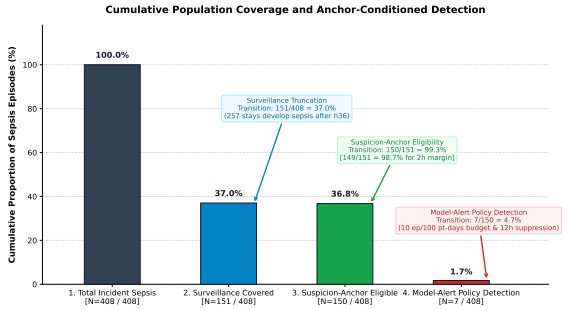


Figure 3: Cumulative Population Coverage and Anchor-Conditioned Detection. Stepwise attrition from 408 incident sepsis episodes through surveillance protocol coverage, retrospective suspicion-anchor eligibility, and model-alert policy detection. *Alt-text:* Four-stage bar chart showing patient opportunity attrition from total incident sepsis (408 patients) to surveillance coverage (151 patients, 37.0%), suspicion-anchor eligibility (150 patients, 36.8%), and timely model-alert policy detection (7 patients, 1.72%).

population coverage and emitted-alert yield. A secondary outcome-stratified hybrid restriction produced small changes in discrimination, but did not represent a uniform prospective eligibility policy. Under a common earliest-prescription-start proxy and declared midnight assignment, no eligible horizon-positive observations remained, preventing discrimination estimation for that restricted task.

2. **Detection Yield Is Constrained by Workload and Prevalence:** AUROC values near 0.64 are modest. At 10 emitted episodes per 100 evaluation-bin patient-days, PPV ranged from 0.98% to 1.72%, corresponding to 58.1–101.8 simulated evaluations per detected patient (NNE); population-level timely yield was 1.72% under the prespecified surveillance schedule.

3. **Surveillance Protocol Coverage Explains Unreached Cases:** Under the evaluated second-day protocol, 63.0% of incident cases develop onset after the monitoring window terminates, preventing multi-hour warning.

4.1 Distinguishing Care-Process Forecasting from Sepsis Anticipation

Forecasting antibiotic initiation and anticipating sepsis represent distinct prediction tasks. When benchmarks predict antibiotic initiation across all patients, they evaluate care-process forecasting (predicting clinician ordering behavior, including prophylactic prescribing in non-septic patients) rather than physiological sepsis anticipation. True early-warning evaluation should retain the physiological sepsis outcome while evaluating eligibility relative to clinical-process anchors; replacing the disease outcome with treatment receipt evaluates an entirely different estimand.

Clinician-vignette studies address when clinicians judge treatment appropriate [11]; our analysis addresses prediction performance relative to operational EHR proxies. These are complementary questions, and proxy precedence does not establish clinical appropriateness. Furthermore, while changes under prescription-proxy restriction were small in this cohort, differences in outcome construction, population selection, timestamp

precision, and clinical-action definition preclude direct replication of Dutta et al.'s order-based sensitivity analysis [13].

Repeated and low-information notifications have been associated with reduced clinician acceptance in other decision-support settings [16]. While we quantify simulated emitted-alert exposure and evaluation yield, our analysis did not measure cognitive overload, alert acceptance, or alarm fatigue directly [16, 17]. Clinical impact also depends on whether alerts are reviewed and acted upon [10]. Our evaluation estimates opportunities and simulated review burden; it does not estimate provider adoption or changes in treatment timing.

4.2 Limitations

The analysis estimates operating characteristics in a selected long-stay ICU cohort rather than clinical effectiveness. Burden-matched cutoffs were derived retrospectively from complete evaluated score trajectories, and identifier-based selection was required for tied boundary scores. Accordingly, the frontier does not represent a prospectively locked policy or guarantee a specified alert rate in online deployment.

Temporal precision also constrains interpretation: MIMIC-III prescription dates cannot be treated as directly observed administration times [20, 23]. Our order-proxy analyses therefore require explicit uncertainty bounds and cannot establish hour-resolved clinical treatment delay. Observed treatment precedence does not establish that infection was unrecognized or that earlier antimicrobial administration would have been appropriate. The study did not measure clinician response, review duration, treatment changes, alarm fatigue, or patient outcomes; NNE therefore represents a simulated evaluation count rather than measured clinical workload. Alarm-reduction interventions may suppress both nonactionable and clinically important signals; our short evaluation interval cannot establish the safety or optimal duration of a refractory policy during sustained surveillance [17]. Finally, our cohorts originate from a single medical center (BIDMC) across EHR system transitions (CareVue to MetaVision); multi-center validation across diverse health systems remains essential.

5 Conclusion

In this restricted, long-stay ICU cohort, observed detection yield depended on surveillance coverage, baseline discrimination, and emitted-alert policies. Evaluation feasibility was highly anchor-dependent: horizon-positive observations were retained before the selected suspicion anchor, but none remained before the earliest prescription-start proxy under midnight assignment, precluding discrimination estimation for that task. Machine learning early-warning evaluations must transparently identify outcome-dependent evaluation rules, report anchor-dependent evaluability, and benchmark under explicit workload constraints.

Ethics Approval and Consent to Participate: This study was conducted using de-identified data from the MIMIC-III (v1.4) database, collected at Beth Israel Deaconess Medical Center (BIDMC). Data collection was approved by the Institutional Review Boards of BIDMC and the Massachusetts Institute of Technology (MIT). The data were de-identified in accordance with HIPAA Safe Harbor standards; informed consent was waived by reviewing IRBs. Access was granted under PhysioNet Credentialed DUA.

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Authors' Contributions: AS conceived the study, developed the analytical framework, implemented pipelines, performed experiments, and drafted the manuscript. SA provided academic supervision, contributed to methodological design, critically revised the manuscript, and approved the final draft.

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Code and Data Availability: Model architectures and frozen weight tensors are archived under commit `b0932ca` (manifest `1.0.0-freeze`); evaluation code under commit `1292103`; denominator recomputation under commit `311cea3`; submission packaging under commit `9b92113`, tag `v1.0.1-freeze`. Preprocessing pipelines are available under MIT License at `https://github.com/alireza/Sepsis-Onset-Timing-Paradox`. De-identified data are available from PhysioNet under MIMIC-III DUA.

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